QUOTIENTS OF SCHEMES BY FINITE LOCALLY FREE FREE GROUP ACTIONS ARE SCHEMES AND TORSORS

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file spaces-properties.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): in this chapter the author's proof environment follows the statement environment as a standalone block; the spec covers the complete environment pair (statement span, proof span) as recorded in the pool candidate.

Editorial note. Occupation scan: combined claim_key + statement_anchor + generic-line scan over all registration files (active and withdrawn) found 13 S10-geom published cards for spaces-properties.tex (prop-occupied.json, 2026-10-14); those 13 labels are excluded from this batch by design. For the 77 free labels: 0 occupied. The registry mentions field (7557 files) and focused-seeds (645) both point to the same S10-geom cards (dual-source check), no other occupation.

Source reference. Original reference 07S2: sources/stacks-chow/tags.txt, source lines 9887–9887. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original proposition 07S6: sources/stacks-chow/tags.txt, source lines 9891–9891. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 01ZY: sources/stacks-chow/tags.txt, source lines 2613–2613. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Source reference. Original reference 049A: sources/stacks-chow/tags.txt, source lines 5573–5573. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let $G \to S$ be a group scheme. Let $X \to S$ be a morphism of schemes. Let $a : G \times_S X \to X$ be an action. Assume that*

1. *$G \to S$ is finite locally free,*
2. *the action a is free,*
3. *$X \to S$ is affine, or quasi-affine, or projective, or quasi-projective, or X is isomorphic to an open subscheme of an affine scheme, or X is isomorphic to an open subscheme of $Proj(A)$ for some graded ring A, or $G \to S$ is radicial.*

Then the fppf quotient sheaf X/G is a scheme and $X \to X/G$ is an fppf G-torsor.

Proof. We first show that X/G is a scheme. Since the action is free the morphism $j = (a, \mathrm{pr}) : G \times_S X \to X \times_S X$ is a monomorphism and hence an equivalence relation, see Groupoids, Lemma 07S2. The maps $s, t : G \times_S X \to X$ are finite locally free as we've assumed that $G \to S$ is finite locally free. To conclude it now

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suffices to prove the last assumption of Proposition 07S6 holds. Since the action of G is over S it suffices to prove that any finite set of points in a fibre of $X \to S$ is contained in an affine open of X. If X is isomorphic to an open subscheme of an affine scheme or isomorphic to an open subscheme of $\mathrm{Proj}(A)$ for some graded ring A this follows from Properties, Lemma 01ZY. If $X \to S$ is affine, or quasi-affine, or projective, or quasi-projective, we may replace S by an affine open and we get back to the case we just dealt with. If $G \to S$ is radicial, then the orbits of points on X under the action of G are singletons and the condition trivially holds. Some details omitted.

To see that $X \to X/G$ is an fppf G-torsor (Groupoids, Definition 049A) we have to show that $G \times_S X \to X \times_{X/G} X$ is an isomorphism and that $X \to X/G$ fppf locally has sections. The second part is clear from the fact that $X \to X/G$ is surjective as a map of fppf sheaves (by construction). The first part follows from the isomorphism $R = U \times_M U$ in the conclusion of Proposition 07S6 (note that $R = G \times_S X$ in our case). $\square$